EmberLink

Linked fire alerts for rural lanes

Proposal and narrative for the Seva First Innovation Challenge 2026

Problem Statement 39 (linked fire alarms for a lane) | Theme 5: Surakshit Bharat, Disaster Resilience | Track A: Product and Technology

Team lead	Sai Shrila, CBIT, Hyderabad
Mentor	Srihari Maddula, Founder and CEO, Eurth Techtronics Pvt Ltd
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1. Introduction

A fire in one house of a thatched lane is a fire in every house of that lane. The people at greatest risk are asleep, the nearest fire station is far away, and by the time a neighbour notices smoke and finds a phone, the next roof has usually caught. EmberLink is a set of small solar-powered detectors, one per house, that warn the whole lane at once and then call for help without anyone having to dial.

When a detector confirms a fire it sounds its own siren and passes the alarm to its neighbours over a direct radio mesh, so every siren in the lane goes off within seconds. One unit in the lane, the gateway, also carries a cellular modem. It sends text messages and places calls to a list of people the Gram Panchayat chooses, and it reports the event to an online dashboard. The lane needs one SIM, not thirty.

We are not starting from a blank page. A working version of the mesh already exists, built on an earlier generation of microcontroller by our mentor's engineering company. It forms and repairs its own network, pushes software updates and device settings to every unit through the mesh, selects its own gateway, and combines a temperature probe with a proprietary optical infrared smoke-sensing method that was designed to ignore cooking smoke. What it lacks is what a village needs before it can adopt it. It is not on current hardware, it cannot dial out, it has no management dashboard or self-test reporting, and it has not run in a real lane with real residents. Those gaps are this project.

The rest of this document sets out the problem, the principles we will hold ourselves to, the design, how we will prove it works, who will run it, what it costs and what could go wrong. Section 4 and Appendix G say exactly which parts exist today and which parts the student team will build.

2. The problem

The 2011 Census recorded that 15.0 per cent of India's 246.7 million households lived under roofs of grass, thatch, bamboo, wood, mud or similar material, and a further 0.6 per cent under plastic or polythene [1]. That is about 37 million households. Pucca construction has spread since 2011 and we do not claim the figure still holds, but it shows the scale of the housing type EmberLink is built for. In our pilot lane we will record the roof and wall material of every house during the field survey rather than rely on a national figure.

The National Crime Records Bureau counted 5,971 fire accidents and 5,888 deaths in 2024 [2]. In 2022, 4,028 of the 7,566 recorded fire accidents happened in residential dwellings [3]. These are police-registered cases. We expect that many fires in kutcha houses are put out by neighbours or burn out without a report, so we read the figures as a floor and not as the full picture.

A lane fire spreads because of how the lane is built. Houses share walls or stand a few feet apart. Roofs are dry organic material. Cooking is often on an open hearth, light may come from a kerosene lamp, and wiring is informal. A spark that lands on thatch next door does not need a second ignition

source. Local fire officers will give us time-to-spread figures for our region during Stage 1. Until then our working assumption is that a lane fire becomes unmanageable within minutes and not hours, which means the first few minutes decide the outcome.

Four things keep existing products from closing that window. A stand-alone smoke alarm wakes one house and leaves its neighbours asleep. An alarm that sounds every time someone lights the chulha gets switched off, or its battery is removed, and then protects nobody. A connected alarm that gives every home its own modem and SIM plan costs more than a Panchayat can sensibly spend across a village. And in nearly every case the call for help still depends on a frightened person finding a phone, remembering a number and explaining where they are.

The gap we are aiming at is the first part of a fire: the minutes when a lane can still be woken, children and animals moved out, and a bucket chain started, and the time before a fire service can arrive. EmberLink wakes every house and starts the call automatically.

3. Tenets

These are the rules we will use to settle design arguments. When two options conflict, the one that follows these wins.

The lane is the unit of protection. A detector that protects one house leaves the neighbouring roofs unwatched. We design, test and price per lane.

No recurring cost for a household. Households do not pay for data plans, subscriptions or batteries. One shared SIM per lane is the only running cost we accept.

A false alarm costs as much as a missed fire. Alarms that cry wolf get unplugged. We will measure false alarms in real kitchens before we call the detector done.

The system reports its own failures. A dead battery discovered during a fire is too late. Every unit tests itself each week and the lane's volunteer gets a message when something needs attention.

Reuse what already works. The mesh, update and profile management exist and have run. Student time goes to what is new. We do not rebuild proven parts to make the project look larger.

The village runs it. A Panchayat volunteer with a phone must be able to test, silence and reset the system. Engineers are for upgrades, not for daily operation.

4. The solution

4.1 What each house gets

Each house has one detector, mounted high on a wall on a non-combustible backing. It has two sensing channels: a proprietary optical infrared smoke chamber and a DS18B20 digital temperature probe. A microcontroller reads both, a 12 V piezo siren makes the noise, and a rechargeable LiFePO4 cell powered by a small solar panel keeps it running through power cuts and cloudy weeks. The default decision rule is that a lane alarm needs both channels to agree. A single channel can raise a local pre-alarm chirp and a log entry, but it does not wake the lane. How the two channels combine, and the thresholds, are held in the unit's profile and will be tuned against real cooking in the pilot.

4.2 How the lane talks to itself

Detectors talk to each other using ESP-NOW, a connectionless protocol that runs on the same 2.4 GHz radio as Wi-Fi but needs no router, access point or SIM [4]. Every detector is also a repeater. An alert frame carries the origin, a sequence number and an authentication code. A neighbour that hears a new alert rebroadcasts it once and drops duplicates, so the alarm floods outward through the lane. The existing implementation maintains the mesh automatically: units find their neighbours, watch link quality and re-route when a unit disappears, so a destroyed or unpowered house does not cut the lane in two.

We authenticate frames in our own software with a key shared across the lane. ESP-NOW has a built-in encryption mode, but the vendor documents only a small number of encrypted peers per device [4],

which does not fit a lane of thirty houses. Settings and software updates travel the other way, from the gateway to every unit through the same mesh, during a declared maintenance window.

To make a small battery last, detectors sleep most of the time and wake for short listening windows. A device in modem sleep cannot receive ESP-NOW traffic [5], so the windows are deliberate and the alert burst lasts longer than one sleep period, which guarantees that every neighbour hears it. We expect each hop to add about a second at worst. We will measure it in Stage 3, and the measurement decides between two schedules described in Appendix A.

4.3 How the lane calls for help

The gateway is a detector with a cellular modem attached, a larger battery and a larger solar panel. The mesh's gateway selection mechanism decides which eligible unit acts as gateway, and the gateway collects the status reports that every unit sends. When it receives a confirmed alarm it sends an SMS to every contact on the lane's list, with the lane name and house number in the local language, then calls the contacts in order until one acknowledges by reply. The same event goes to the dashboard over the modem's data connection. Contacts are the Panchayat office, lane volunteers and the nearest fire station, agreed with each of them beforehand.

Authorised phone numbers can also send commands by SMS: STATUS returns the node count, battery levels and health, TEST sounds a short lane-wide test, RESET clears a latched alarm after an incident, and SILENCE mutes the sirens for ten minutes during controlled burning while the sensors keep logging. If the lane has no cellular signal the mesh still wakes every house, which is why we check coverage at the gateway position, for each operator, before choosing a lane.

4.4 What we removed, and why

An early design iteration paired the mesh with a LoRa radio in every detector and in the gateway. We removed it from both. Houses in a lane are tens of metres apart, which the existing mesh already covers. A second radio adds cost, a second antenna, a second protocol to maintain and extra current drawn from a small cell. The cellular modem already covers the long distance to the outside world. If the survey finds a lane too long or too obstructed for ESP-NOW, we add plain repeater units running the same firmware, not a second radio system.

4.5 What exists, and what we will build

Area	Already working (prior implementation)	Built in this project
Radio mesh	ESP-NOW mesh on ESP8266 with automatic maintenance of neighbours and routes	Port to ESP32; confirm range and per-hop delay; low-power listening schedule; authenticated alert frames
Updates and settings	Software updates and device profiles pushed through the mesh	Dashboard screens for profiles and update campaigns; staged rollout and rollback; battery check before an update
Gateway	Gateway selection and aggregation of node reports	Cellular bridge: SMS, calls, cloud posting, SMS commands, retry and escalation
Sensing	DS18B20 temperature probe; proprietary optical IR smoke sensing that rejects false alarms	Threshold calibration in real kitchens; dust-drift compensation; per-house-type profiles
Power	Battery-powered node hardware	Rev B board: LiFePO4 solar charger, siren boost stage, weekly battery pulse test
Operations	Not yet built	Commissioning walk test, lane dashboard, weekly health message, training, handover manual, pilot data

The move from ESP8266 to ESP32 is part of the work. It gives us an external antenna connector for difficult lanes, hardware acceleration for the authentication code, two cores, better sleep modes and more memory for safe over-the-air updates. The ESP8266 build stays as the reference, and the ESP32 build must reproduce its behaviour on the same test script before we add anything new.

5. Methodology and approach

The work runs in five stages. The dates fit the Seva First Innovation Challenge calendar: submissions close on 24 October, screening begins on 25 October, the exhibition runs from 30 November to 15 December, and the national finale is in New Delhi from 10 to 17 December [6][7]. Our dates after 24 October move if the official calendar does.

Stage 1, field survey and baseline (25 October to 1 November). With the Gram Panchayat we choose a lane and walk it. We record house positions, roof and wall materials, how far apart the houses are, how each house cooks and lights, and signal strength of the main mobile operators at candidate gateway positions. We interview residents and the local fire officer, and we measure the existing ESP8266 build for battery current, hop delay and false alarms, so every later number has a baseline.

Stage 2, port and build (25 October to 15 November). The firmware is ported to ESP32 behind a thin hardware layer, so sensing and mesh code do not change when the board does. In parallel we build the Rev B board, the cellular bridge, the dashboard and the weekly self-test. If the Rev B board arrives late, the pilot runs on ESP32 development modules wired to the existing sensor boards.

Stage 3, bench and campus lane test (9 to 16 November). Ten units and a gateway are laid out as a mock lane on campus. We run the tests in Appendix C, except the real-kitchen tests. Smoke tests use a chamber, supervised by faculty. We light no open flame in or near any house at any stage.

Stage 4, live lane pilot (17 to 27 November). Subject to Panchayat permission, we install the gateway and ten detectors in the surveyed lane. We calibrate the dust baseline and thresholds against real cooking, test every dispatch route with the contacts, hold a drill with residents, and train the volunteer. False alarms are logged and classified as they happen.

Stage 5, handover and exhibition (28 November to 15 December). The Panchayat receives the installation, a manual in the local language and the spare units. At the exhibition we run a live lane. A judge can trigger a test alarm, watch the sirens cascade with a timer, receive the SMS on a phone, switch off a unit and watch the mesh route around it.

We will judge success against the targets below, each measurable in the pilot, with methods in Appendix C. We will report the results whatever they are.

#	Measure	Target	How we measure it
M1	False alarms	No more than 1 in 100 unit-days, in homes that cook daily	10 units for 14 days, 140 unit-days; every alarm logged and classified
M2	Detection	Every test event detected; time compared with a certified reference	At least 30 supervised chamber tests, smouldering and flaming
M3	Lane alert time	All sirens sounding within 5 s of confirmation; about 1 s per hop at worst	At least 200 triggered alerts on a ten-unit lane with at least 3 hops
M4	Dispatch	SMS reaches every listed contact within 30 s in at least 95% of trials	50 trials on the pilot SIM, delivery checked on test phones
M5	Self-healing	Reports reach the gateway within 60 s after any one unit is switched off	Each unit removed in turn
M6	Battery	Average current at most 4 mA, which supports 14 days without sun	Logged for 7 days with a current meter, then extrapolated
M7	Remote update	Update reaches every unit with no site visit; a failed update rolls back	Push to 10 units, force one failure
M8	Self-test	All injected faults flagged in the weekly report	Inject dust, a weak battery and a blocked link on 3 units each

6. Team and operating model

Role	Person	Responsibility
Team lead and software lead	Sai Shrila	Cellular bridge, dashboard, self-test analytics, pilot data and documentation; runs the field survey
Mentor and technical guide	Srihari Maddula	Founder and CEO of Eurth Techtronics Pvt Ltd, a Hyderabad electronics design company with more than 26 delivered products. Provides the existing mesh implementation, design review and access to his engineering team
Firmware and hardware support	Eurth Techtronics engineers [names to confirm]	ESP32 port, Rev B board, test fixtures
Team members	[names to confirm]	Assembly, field survey, community training, testing
Faculty guide	[name, CBIT]	Safety review of battery and enclosure design; supervision of smoke tests
Community partner	[Gram Panchayat to confirm]	Pilot lane, nominated lane volunteer and deputy, contact list

After handover the Panchayat nominates a lane volunteer and a deputy. We train them in about an hour to read the weekly health message, send the four SMS commands, swap a failed unit from the two spares the Panchayat keeps, and call us for anything else. Software updates and profile changes are pushed remotely by our team during the pilot and by whoever the Panchayat appoints afterwards.

Funding for a lane at the cost in Section 7 is a decision for the Panchayat and its funders. Routes we will explore are Gram Panchayat funds, state disaster-management funds, corporate social responsibility programmes and a staged household contribution. We do not propose a subscription. The pilot is paid for by the team.

7. Cost

A detector unit costs ₹1,450 and the gateway, which includes a detector, costs ₹3,110. A lane of thirty houses needs twenty-nine detectors and one gateway, which comes to ₹45,160, or ₹1,505 per house. The itemised bill is in Appendix B. These prices carry forward our earlier cost sheet, adjusted for the changes in this design, and are planning estimates that Stage 2 replaces with supplier quotations.

The running cost is one prepaid SIM per lane. At an assumed ₹250 a month, that is about ₹8 per house per month. A design that gave every house its own modem would add ₹27,550 of modem hardware per lane and twenty-nine more SIM plans, about ₹7,250 a month at the same assumed price. That difference is the reason the design has one gateway and a mesh.

Item	Estimate
Detector unit, per house	₹1,450
Gateway unit, one per lane (includes a detector)	₹3,110
Lane of 30 houses (29 detectors and 1 gateway)	₹45,160
Lane cost per house	₹1,505
Running cost per lane per month (assumed SIM plan)	₹250
Pilot: 10 detectors and 1 gateway, 2 spare detectors, test fixtures, Rev B prototype run, travel, SIM, 15% contingency	₹36,237

8. Risks and open issues

Appendix E has the full register. Six matter most.

Cellular voice and SMS on a 4G-only modem. The SIM7670G is a 4G Cat-1 module, and the vendor's feature list for the HAT names data protocols and positioning but does not state voice calling [8]. On Indian networks, voice and SMS over 4G depend on the operator provisioning the module for VoLTE. We will test the target operators' SIMs in the first week. If voice does not work, the gateway posts the event over its data connection and the dashboard service places the calls and SMS through a telephony provider. The detectors and mesh do not change.

Power in a thatched house. A lithium cell in a house made of dry thatch is a hazard if the charger or enclosure is wrong. Charging uses a LiFePO4 profile, the cell has a protection circuit and fuse, the enclosure is flame retardant, and the unit mounts on a non-combustible backing. The faculty guide reviews the design before anything is installed in a home.

Delay with sleeping radios. The one-second-per-hop figure is an estimate. The Stage 3 test decides whether we use unsynchronised listening windows or windows synchronised by a beacon from the gateway, and if neither meets M3 we add always-powered repeaters at a few points in the lane.

False alarms in real kitchens. The optical method and temperature channel were designed to reject cooking smoke, but we have no data from this lane. M1 exists to test that. Thresholds are tuned per house type during the pilot.

Schedule. Panchayat permission and the Rev B board are the two items outside our control. Both have a fallback: the campus mock lane and development modules on the existing sensor boards.

What this is and is not. EmberLink is an early-warning aid. It is not a certified fire alarm and does not replace the fire service. We will say so on the unit, in the manual and in training, and the drill will include calling the fire service directly.

9. What we ask

We ask the Challenge to advance EmberLink to the exhibition stage so we can complete the pilot. We would also welcome an introduction to a Gram Panchayat willing to host a lane, advice from a fire-service officer on test procedure, and access to a test space for the campus lane.

If the pilot meets M1 to M8, the Panchayat in the pilot village will have a working lane that costs about ₹1,500 per house, needs no subscription from households, and tells its own volunteer when it needs attention. If a target is missed, we will say which one and what we would change.

Appendix A. Technical specification

This appendix is written for engineers and evaluators. Part names are given here and not in the narrative. Where a value is a design target and not a measurement, it says so.

A.1 System overview

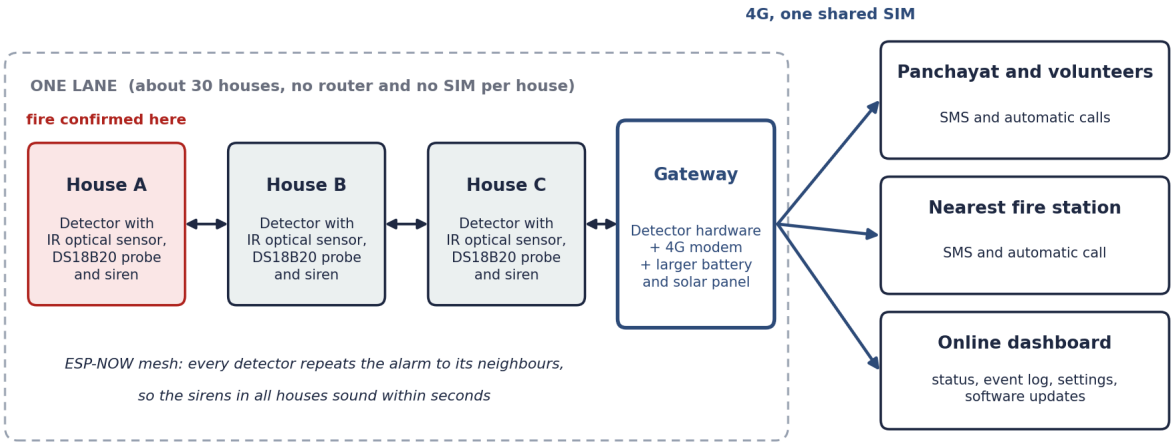


Figure A1. One lane: detectors linked by ESP-NOW, one gateway with a 4G link to the Panchayat, the fire station and the dashboard.

A.2 Detector unit

Subunit	Component	Specification and purpose
Processor and radio	ESP32-WROOM-32U	Dual-core, 2.4 GHz radio used for ESP-NOW, connector for an external antenna. No second radio.
Smoke sensing	Optical IR chamber (proprietary)	Existing design from the mentor's organisation, built to reject cooking smoke. Evaluators can see its behaviour in the tests; circuit-level detail is available under NDA.
Temperature	DS18B20 digital probe	One-wire, specified at 0.5 °C accuracy around room temperature. Read at 10-bit resolution (about 190 ms per conversion) to keep sampling energy low. Exposed to room air through a vent.
Audible warning	12 V piezo siren with boost stage	Target at least 85 dB at 1 m, measured in Stage 3 at bed distance. Driven by a MOSFET from a boost converter, since the cell is 3.2 V.
Battery	LiFePO4 18650, 3.2 V, 2000 mAh	Protection circuit and fuse. LiFePO4 chosen for thermal stability.
Charging	LiFePO4 solar-input charger	Set to the 3.6 V LiFePO4 profile. Must not be a 4.2 V lithium-ion charger.
Solar	5 V, 2 W panel	Keeps the cell topped up through monsoon weeks.
Telemetry	ADC1 inputs and a load MOSFET	Battery and panel voltage through 100 k / 100 k dividers; a 20 ms load pulse for the weekly internal-resistance test.
Enclosure	Flame-retardant ABS, IP65	Mounted on a non-combustible backing.

A.3 Gateway additions

Addition	Specification and purpose
SIM7670G 4G HAT	Serial AT-command interface to the microcontroller. SMS, calls (subject to operator VoLTE provisioning) and data. Position can come from the HAT's GNSS or from fixed coordinates recorded at installation.

Addition	Specification and purpose
Supply for the modem	A boost converter to the voltage the HAT requires, sized for the modem's transmit peaks. A single LiFePO4 cell at 3.2 V nominal is below the supply range of typical Cat-1 modules, so it cannot feed the modem directly.
Battery and panel	6000 mAh LiFePO4 pack and a 6 W panel.
Antenna	External 4G antenna on a short lead, placed after the cellular survey.

A.4 Reference pin assignment

This is the reference assignment for Rev B. The final mapping follows the Rev B schematic and the existing sensor board. All analogue inputs are on ADC1, because ADC2 cannot be read while the radio is active.

Function	GPIO	Type	Note
Optical IR receiver output	34	Analogue in, ADC1	Input-only pin
Optical IR emitter drive	26	Digital out	Short pulses to save energy
DS18B20 data	27	One-wire	4.7 k pull-up to 3.3 V
Siren drive (MOSFET and boost enable)	25	PWM out	
Battery voltage	32	Analogue in, ADC1	100 k / 100 k divider
Panel voltage	33	Analogue in, ADC1	100 k / 100 k divider
Load-pulse MOSFET	14	Digital out	Weekly internal-resistance test
Test button	13	Digital in, RTC wake	Local test and silence
Status LED	2	Digital out	Strapping pin; keep high-impedance at boot
Modem UART RX / TX (gateway only)	16 / 17	UART2	3.3 V logic
Modem PWRKEY / RESET (gateway only)	4 / 5	Open-drain out	PWRKEY held low for about 1.2 s to power-cycle; GPIO5 is a strapping pin

A.5 Alert and message content

The existing mesh protocol is retained. The list below is the content the dispatch layer needs from it. Names follow our documentation, and Stage 2 maps them to the existing implementation. An ESP-NOW frame holds at most 250 bytes [4], which is ample for all of these except software update data, which is split into chunks.

Message	Content
ALERT	Origin unit, house number from its profile, level (pre-alarm or confirmed), sensor snapshot, sequence number, authentication code. Rebroadcast once by each neighbour; duplicates dropped by origin and sequence.
STATUS	Battery voltage, panel voltage, optical baseline and drift, last self-test result, link quality to neighbours, profile and software version. Aggregated by the gateway.
PROFILE	Thresholds, sampling interval, siren pattern, silence window, house number. Versioned; the unit keeps the previous version for rollback.
UPDATE	Software chunks with a bitmap of missing pieces. Sent only in a maintenance window and only to units above a battery threshold.

A.6 Power plan

These figures are planning estimates from datasheet-class values. Test T6 replaces them with measurements.

Contribution to average current (detector)	Average (mA)	Basis
Sleep floor (light sleep, regulators, sensors idle)	0.8	About 0.8 mA light-sleep figure
Listening windows: about 100 mA for 20 ms every second	2.0	2% duty
Sensing: DS18B20 conversion and IR pulse every 2 s	0.4	Short, low-current
Heartbeat and mesh upkeep	0.2	Brief transmissions
Total	**3.4**	Target at most 4 mA (M6)

With 80% of 2000 mAh usable, 3.4 mA lasts about 19.6 days without any sun, against a requirement of 14. The panel gives the cell far more energy in a day than the unit uses, so its job is to survive weeks of cloud. For the gateway, the node load plus modem standby is a planning figure of about 40 mA at the battery, which gives about 5 days from a 6000 mAh pack without sun. Our gateway target is at least 5 days, with the 6 W panel recovering it.

A.7 Mesh schedule options

Option 1 uses unsynchronised listening windows, one second apart. A detecting unit sends repeated frames for slightly more than one second so every neighbour's window overlaps, and each hop adds up to about a second. It needs no coordination and costs nothing in complexity. Option 2 has the gateway send a periodic time beacon and all units wake in the same short window, so alerts cross hops in milliseconds, at the cost of beacon energy and drift correction. Stage 3 measures both on the same lane and we adopt the simpler one that meets M3.

A.8 Gateway bridge

The bridge is a state machine over the serial AT interface. It powers the modem, checks network registration and signal, and keeps a queue of outgoing events that survives resets. For each event it sends the SMS to every contact, retries on failure with back-off, then calls contacts in order, stopping when one replies OK. It posts the same event to the dashboard. SMS text is stored in the local language. Telugu or Hindi text is sent as Unicode, which limits each SMS segment to 70 characters, so messages are short and fixed. Incoming SMS is accepted only from the allow-list of authorised numbers.

Command	Action	Reply
STATUS	Report units seen, battery levels, last self-test, open maintenance flags	One consolidated SMS
TEST	Sound a short low-volume lane-wide test	Count of units that responded
RESET	Clear latched alarm state after an incident	Confirmation
SILENCE	Mute sirens for 10 minutes; sensing and logging continue	Confirmation and time remaining
OK	Acknowledge an alarm and stop further calls	None

A.9 Dashboard and cloud service

The dashboard is the main new software component. It has an ingestion endpoint that the gateway posts to over HTTPS or MQTT, a database, and a web interface. The interface shows the lane as a map of houses with each unit's state (normal, pre-alarm, alarm, needs attention, offline), an event log with filters and export, a profile editor with version history, an update manager that rolls software out in stages and rolls back on failure, and a contacts editor. There are three roles: Panchayat administrator, lane volunteer (read, test, silence) and engineer. Each change is logged with the user and time. The service holds only unit telemetry and contact numbers, with no audio, video or personal profiles. Technology choices are left to the software lead, with a preference for widely used open components so the Panchayat is never tied to one vendor.

A.10 Weekly self-test

Every Sunday at 02:00 each unit wakes and runs four checks. It compares the optical chamber's background reading with its baseline and updates the baseline with an exponential moving average. If drift exceeds 35% of the dynamic range it raises a maintenance flag. It switches a known load across the cell for 20 ms and compares loaded and unloaded voltage to estimate internal resistance, which shows an ageing cell before it fails. It exchanges a round-trip frame with its neighbours to record link quality. Finally it sends its results to the gateway, which sends one consolidated message to the lane volunteer. These checks double as the evidence for M8.

Appendix B. Bill of materials and cost model

All prices are planning estimates in rupees. They carry forward the earlier cost sheet and add the parts this design needs: the temperature probe, the siren boost stage, the modem supply and the larger gateway battery and panel. Stage 2 replaces every line with a dated supplier quotation.

B.1 Detector unit

#	Item	Qty	Unit cost	Line cost
1	ESP32-WROOM-32U module (external antenna connector)	1	₹280	₹280
2	Optical IR smoke chamber assembly (existing proprietary design)	1	₹190	₹190
3	DS18B20 temperature probe	1	₹50	₹50
4	12 V piezo siren	1	₹45	₹45
5	Boost converter, battery to 12 V, for the siren	1	₹50	₹50
6	LiFePO4 18650 cell, 3.2 V, 2000 mAh, with protection	1	₹180	₹180
7	Solar-input charger for LiFePO4 (3.6 V profile)	1	₹85	₹85
8	5 V, 2 W solar panel	1	₹160	₹160
9	Custom PCB, enclosure and hardware	1	₹240	₹240
10	Passives, MOSFETs, LEDs, connectors	1	₹170	₹170
	Detector unit total			**₹1,450**

B.2 Gateway additions

#	Item	Qty	Unit cost	Line cost
1	SIM7670G 4G HAT	1	₹950	₹950
2	External 4G antenna	1	₹120	₹120
3	Battery upgrade to 6000 mAh pack (net of the 2000 mAh cell it replaces)	1	₹150	₹150
4	Boost converter for the modem supply (2 A class)	1	₹150	₹150
5	Larger IP65 enclosure (net)	1	₹100	₹100
6	Solar panel upgrade to 6 W (net of the 2 W panel it replaces)	1	₹190	₹190
	Gateway additions total			**₹1,660**
	Gateway unit total (detector plus additions)			**₹3,110**

B.3 Lane and pilot totals

Item	Estimate
Lane of 30 houses: 29 detectors and 1 gateway	₹45,160
Per house	₹1,505
Pilot: 10 detectors	₹14,500
Pilot: 1 gateway	₹3,110
Pilot: 2 spare detectors	₹2,900
Pilot: Rev B board prototype run and assembly	₹3,000
Pilot: Test fixtures: smoke test chamber, reference detector, current logger	₹4,000
Pilot: Travel, installation material and drill logistics	₹3,000
Pilot: Prepaid SIM and data for the pilot period	₹1,000
Pilot subtotal	₹31,510

Item	Estimate
Contingency, 15%	₹4,727
Pilot total	**₹36,237**

B.4 Running cost and comparison

One prepaid SIM per lane at an assumed ₹250 a month. Dashboard hosting is a small fixed cost that is shared across all lanes on the platform and, during the pilot, is carried by the team. A per-house modem design would add 29 modems at ₹950 each (₹27,550) and 29 SIM plans (₹7,250 a month at the same price) to a thirty-house lane.

Appendix C. Test and validation plan

Tests T1 to T8 correspond to targets M1 to M8 in Section 5. T9 and T10 cover the enclosure and the port. Safety rule for every test: no open flame in or near any house. Smoke tests are run outdoors in a chamber under faculty supervision with extinguishing equipment on hand.

Test	Objective	Method	Pass rule
T1	False alarms	10 units in occupied homes for 14 days. Each alarm and pre-alarm is logged with time, sensor values and what was happening in the kitchen, and classed as true, nuisance or fault.	No more than 1 confirmed alarm in 140 unit-days
T2	Detection	At least 30 events in an outdoor chamber with smouldering and flaming sources, with a certified detector alongside as a reference. Record time to alarm for both.	All events detected; times reported
T3	Lane alert time	Ten-unit campus lane with at least 3 hops. A logging unit timestamps each siren trigger. At least 200 alerts, with the origin rotated across the units.	All sirens within 5 s in at least 99% of alerts
T4	Dispatch	50 trials on the pilot SIM. A confirmed alarm is injected at the gateway and delivery is timed on test phones. Repeat on each candidate operator.	At least 95% of SMS to all contacts within 30 s
T5	Self-healing	Switch off each unit in turn during traffic and time the recovery of reports to the gateway.	Recovery within 60 s for every unit
T6	Power	Log current with a meter for 7 days on three units and on the gateway; fit the measured profile to the battery and extrapolate.	Detector at most 4 mA; gateway at least 5 days without sun
T7	Remote update	Push a new build and profile to ten units, force one failure by interrupting the transfer, and confirm rollback.	All good units update; the failed unit rolls back
T8	Self-test	On three units each, inject chamber dust, a weak cell and a blocked link, then run the weekly test.	Every injected fault in the weekly report
T9	Enclosure and charging	Sun and heat soak with the unit charging, spray test for ingress, and a check that nothing in the enclosure exceeds its rating.	No damage; no overheating; charger stops at set limits
T10	Port equivalence	Run the same scripted scenario on the ESP8266 build and the ESP32 build, and compare alerts, timing and battery current.	ESP32 matches or beats the baseline on every item

Appendix D. Schedule and deliverables

Work	Dates	Deliverable
Submission	19 to 24 October	Proposal and concept note submitted; parts ordered; pilot lane request made to the Panchayat
Field survey and baseline	25 October to 1 November	Lane map, materials record, signal survey, baseline numbers for the existing build
ESP32 port	25 October to 8 November	ESP32 firmware passes T10
Rev B board	Order by 30 October	Prototype boards; fallback is development modules on existing sensor boards
Gateway bridge and dashboard v1	2 to 15 November	SMS, call, cloud posting, commands; dashboard with map, log and profiles
Campus lane tests	9 to 16 November	Results for T2 to T9; decision between mesh schedules
Live lane pilot	17 to 27 November	10 units and a gateway installed; T1 data; drill; trained volunteer

Work	Dates	Deliverable
Handover and exhibition	28 November to 15 December	Manual in local language; spare units; exhibition lane; results poster

The SFIC calendar governs all dates after 24 October. These are our plan against it.

Appendix E. Risk register

#	Risk	Likelihood	Impact	Mitigation
R1	Voice or SMS does not work on the chosen 4G modem and operator	Medium	High	Test each operator's SIM in week 1; add the data path with a telephony service as fallback; choose the operator per site
R2	Battery or charger becomes an ignition source in a thatched house	Low	High	LiFePO4 charging profile, protection circuit, fuse, flame-retardant enclosure, non-combustible mounting, faculty review before install
R3	Delay across hops exceeds target because radios sleep	Medium	Medium	Measure both schedule options; add always-powered repeaters at a few points
R4	False alarms from cooking smoke or humidity	Medium	High	Two-channel confirmation; thresholds tuned per house type; dust drift compensation; local SILENCE
R5	Siren too quiet at bed distance	Medium	Medium	Measure in Stage 3; adjust boost voltage or siren part; test with doors closed
R6	ESP32 port slips	Medium	Medium	Hardware layer; ESP8266 build as reference; mentor engineers time-boxed to the port
R7	No pilot lane in time	Medium	High	Approach two Panchayats now; campus mock lane as fallback; report honestly which tests were not run in a village
R8	Misuse of SMS commands	Low	Medium	Allow-list of numbers; rate limit; every command acknowledged and logged
R9	Residents think the system replaces the fire service	Medium	High	Clear labelling, training, and a drill that includes calling the fire service directly
R10	Dust, insects and moisture foul the optical chamber	High	Medium	Fine mesh, weekly self-test and drift flag, cleaning step in the volunteer's routine
R11	Cellular signal is poor at the lane	Medium	Medium	Survey before choosing the lane; external antenna; the mesh still wakes the lane without a signal
R12	Contact numbers and event data misused	Low	Medium	Minimal data; role-based access; audit log; no audio or video collected

Appendix F. Questions we expect

Why not use a commercial smoke alarm?

A commercial alarm protects one house and sounds only there. Linked and connected versions assume mains power or a modem and SIM plan per home. EmberLink is designed around a shared mesh, solar power and one SIM per lane, so the cost can be borne by a Panchayat.

Why did you drop LoRa?

Houses in a lane are close enough for ESP-NOW, and the 4G modem already covers the long link to the outside. A second radio would add cost, an antenna, a second protocol and battery drain, with no benefit inside a lane. If a lane is too long, we add repeater units on the same radio.

How can an alert cross 30 houses in seconds?

Each detector repeats the alert once. With listening windows about a second apart, each hop adds up to about a second at worst, so a lane reaches its far end in seconds. We state that as a target and measure it in T3. Windows synchronised by the gateway can make it faster if needed.

What happens if the gateway fails or the cell network is down?

Local alarms do not depend on the gateway or the network, so every house still wakes. The gateway selection mechanism can promote another eligible unit, and events queue on the gateway until the network returns. The weekly health message tells the volunteer if the gateway has been offline.

What happens if a house has no unit?

Its neighbours' sirens still sound, which is the point of protecting the lane. We will report participation in the pilot, and we expect a lane to need broad participation to work as designed.

Who pays for the SIM and how is it kept active?

The Panchayat pays for one prepaid SIM per lane. The weekly health message doubles as activity on the SIM, and the dashboard flags a SIM that is close to expiry.

Is it certified?

No. It is an early-warning aid and will be labelled as one. Certification is a later step if a Panchayat or funder wants a certified product.

What can judges verify in the exhibition?

A live lane with working sirens, a timed alert cascade, an SMS arriving on a phone, a unit switched off with the mesh routing around it, a weekly self-test with an injected fault, and the pilot data.

What exactly did the students build?

See Appendix G. In short, the students own the gateway bridge, dashboard, self-test analytics, field survey, pilot, training and documentation. The mesh, update and profile machinery and the sensing method come from the mentor's earlier implementation, and the port to ESP32 is done jointly with his engineers.

What stays proprietary?

The optical IR sensing method stays with the organisation that developed it. Everything the students write for this project, including the dashboard, bridge and installation guide, is theirs, and we are open to releasing the installation guide and dashboard for Panchayat use.

Appendix G. Prior work and student contribution

The Challenge rewards student innovation. We want to be exact about what exists and what is new, so evaluators can judge the project on the right basis.

G.1 Existing, from the mentor's organisation

A working mesh fire-alerting implementation on ESP8266 with ESP-NOW that maintains the mesh automatically, pushes software updates and device profiles through the mesh, selects a gateway and aggregates node reports. Node hardware with a battery, a DS18B20 temperature probe and a proprietary optical IR smoke-sensing method that rejects false alarms. We will record its measured baseline in Stage 1 and state it in the exhibition material.

G.2 New in this project, led by the students

#	Work package	Output
1	Cellular gateway bridge: SMS, calls, queue, retry, escalation, acknowledgement, local-language messages, SMS command parser	Firmware module; trial results (T4)
2	Dashboard and cloud service: telemetry ingestion, map, event log, profile editor, update manager, roles, audit log	Working service; screens and demo
3	Weekly self-test analytics: drift, battery resistance trend, link quality, consolidated message	Firmware and dashboard feature; T8 results
4	Field survey, pilot lane installation, drill and volunteer training	Survey report; T1 data; training record
5	Installation guide and handover manual in the local language	Printed manual and one-page quick card

G.3 New in this project, done jointly with mentor's engineers

#	Work package	Output
6	ESP32 port behind a hardware layer, with equivalence to the existing build	Firmware; T10 results
7	Rev B board: LiFePO4 solar charging, siren boost, modem supply on the gateway	Boards; T6 and T9 results
8	Mesh schedule decision between unsynchronised and beacon-synchronised listening	Measurement report (T3)

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